Tidal gravity across interrupted optical clock comparisons

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Optical clocks probe gravitational potential through its effect on the passage of time.[1-3] Establishing this capability under interrupted operation requires separating tidal response from clock and transfer variations. Here we analyse independently acquired Shanghai–Wuhan strontium–ytterbium comparisons over a 1,350-km fibre route.[4] Six periods in 2026 provide 20.1 days of retained sample time across a 231-day span. With ten-day calendar baselines, the scale of the predicted tide is 0.88, with a working standard error of 0.10 and a period-resampling 95% acceptance interval of 0.51–1.42. Amplitudes learned from other periods improve the held-out residuals in all six periods. Allowing independent solar-day variations leaves a lunar semidiurnal scale of 0.89, but period resampling no longer excludes zero at the 5% threshold. A simultaneous round-trip record measures sensitivity to one transfer channel. Seasonal solar-tide sensitivity remains more than three orders of magnitude above the predicted differential signal. Together with contemporaneous independent network measurements,[5] these results demonstrate how interrupted clock comparisons test tidal response, while distinguishing accumulated measurement capability from the calibration needed to identify slower gravitational changes.

The gravitational redshift makes a clock's rate a local probe of gravitational potential. Optical-clock development has brought this principle into precision measurement,[6] from laboratory redshift tests[7] to resolving a potential gradient within a millimetre atomic sample.[8] Coherent networks[1,9] and transportable clocks[2] extend the comparison between locations. Static geodetic calibration and time-variable potential require distinct treatment:[10,11] the Moon and Sun, deformation of the Earth and redistribution of mass produce temporal signals superposed on the clocks' intrinsic frequency ratio and systematic shifts.

For an intermittently operated comparison, elapsed calendar time and sensitivity to a tidal component are different quantities. Within-interval variation may constrain a signal even when frequency levels cannot be transferred across every interruption. Conversely, a long calendar span need not constrain a slow signal when clock corrections or instrumental changes remain uncertain. Daily environmental variations create a further distinction between a response to the full predicted tide and the lunar semidiurnal response after allowing solar-day terms.

The contemporaneous European measurements of Stahl and colleagues track tidal potential variations with a clock network and Kalman estimation.[5] Our independently acquired Shanghai–Wuhan comparisons test the response across separated observation periods. Our earlier ratio preprint reported a tidal correlation in the summer record.[4] Here we establish a whole-template benchmark, separate lunar and solar-day variation, and examine the different calibration requirements of longer-period sensing. The commissioning status and changed processing of the 2025 observations motivate their separate analysis; they do not enter the common 2026 amplitude. Table 1 distinguishes the overlapping data and statistical targets of earlier and present analyses.

Clocks, link and observation periods

The comparison uses optical lattice clocks based on ^{87}Sr in Shanghai and ^{171}Yb in Wuhan. The associated Sr1 evaluation reports a systematic uncertainty of 9.2×10^{-19}, with chamber thermal characterization and measurements of lattice, density and magnetic shifts.[12] The Yb evaluation reports 1.3×10^{-18}, using an in-vacuum lattice cavity and blackbody shield, and differential comparisons of two clocks.[13] These evaluations describe specified apparatus configurations; they do not provide a complete uncertainty budget for every observation period in the present record.

The 1,350-km route uses commercial fibre, relay stations and bidirectional optical amplification to transfer an optical carrier from Shanghai to Wuhan.[4] Digital phase recording and bias compensation have also been demonstrated on a distinct 2,067-km field link,[14] alongside other long-distance laser-network comparisons.[15] Those results establish technical context. Our transfer check uses the actual simultaneous monitor below rather than importing another route's instability as an error bound.

The principal record contains 1,737,073 retained source measurements in 2,905 non-overlapping 600-s averages, drawn from 49 effective intervals and six observation periods between 7 January and 26 August 2026. Measurements occur on 35 BJT calendar days over a 231-day span. At the nominal one-second gate, the retained source count represents 20.1 days of accumulated sample time. This technical

exposure is distinct from elapsed time and from the effective number of independent observations under correlated noise. The periods are January, April, May, June–July, and early and late August. Ratios follow the recorded frequency chain and correction ledger, retaining the measured interval levels relative to one numerical reference (Methods). This coordinate is not a new absolute-ratio determination.

The 2025 records were acquired during clock commissioning, with different frequency conversion and processing and incompletely resolved offsets, according to the experimental team. We analyse September and October separately using one stage intercept or interval intercepts. They are neither joined to each other nor pooled with 2026. This scope follows experimental clarification, not a choice of the year giving the preferred tidal amplitude.

Response across observation periods

We first fit a signed scale A_T multiplying the entire predicted fractional-frequency tide $T(t)$,

$$y(t) = A_T T(t) + \gamma_{b(t)} + \epsilon(t),$$

where $b(t)$ assigns observations to fixed ten-day BJT calendar blocks. Ten blocks contain data. The amplitude and baselines are estimated jointly; measured points are not individually shifted or aligned to the prediction. These baselines are used for the short-period comparison, not for seasonal inference.

The result is $A_T = 0.8836$, with residual-scaled least-squares, delete-day and delete-period standard errors of 0.0882, 0.1002 and 0.0971. We summarize it as 0.88 ± 0.10, retaining the largest standard error. Five-day blocks give 0.886 ± 0.101. Adding an independent linear drift within every occupied block gives 0.878 ± 0.101 for ten days and 0.868 ± 0.112 for five days. The ten-day change is -0.006 ± 0.030 under paired day deletions. These are correlated sensitivity estimates on the same observations.

Figure 1a shows the 48-h window with the largest retained source count, selected by source count rather than tidal agreement. The rule is reproducible but was not prospectively registered before all exploratory analyses. Points retain their measured levels after subtracting one common plotting reference. The displayed template amplitude, 0.830, is estimated from the other five periods; only the prescribed calendar-block baselines are fitted within the held-out period. Three-hour averages aid visualization without bridging gaps.

The actual-time harmonic diagnostic uses the same nuisance projection before and after template subtraction. At the M2 frequency, variance reduction per observation falls from 13.02×10^{-36} to 0.12×10^{-36}, a 99.1% decrease (Fig. 1b). This is a phase-sensitive, in-sample subtraction diagnostic, not an independent discovery test. In leave-period-out comparisons, training amplitudes range from 0.830 to 0.965. Applying them to held-out periods, with their nuisance baselines fitted locally, reduces residual sums of squares in all six periods, by 3.26% when pooled (Fig. 1c). The folds share training data and are not independent experimental replications. Accumulated sample time provides an exposure measure, but not a universal sensitivity law. Along the actual time-ordered record, eight nominal sample days give whole-template and lunar working errors of 0.114 and 1.025, respectively; at 20.1 days these are 0.100 and 0.282. A fixed independent-bin noise benchmark also shows that lunar separation gains depend on changing phase coverage and nuisance design, beyond exposure alone (Methods, Table 4).

Lunar and solar-day variation

The whole-template model fixes relative tidal amplitudes and phases. Allowing independent sine and cosine terms at exactly 24 h and 12 h within each ten-day block gives $A_T = 0.79 \pm 0.28$, and the M2-frequency diagnostic decreases by 93.8%. Five-day blocks give 0.92 ± 0.38. The daily terms permit instrumental or environmental variations without identifying their origin.

To isolate the lunar semidiurnal response, we fit a common M2+N2 scale while assigning separate scales to the near-12-h and diurnal tidal groups. With the free daily terms retained, $A_L = 0.8909$, with least-squares, delete-day and delete-period standard errors of 0.2344, 0.2824 and 0.1881. The five-day result is 0.89 ± 0.43. Within-block linear drifts give 0.89 ± 0.29 for ten days and 0.82 ± 0.56 for five days. The lunar response remains compatible with the predicted scale under these specified controls (Fig. 2a).

A common lunar quadrature tests phase consistency without adjusting observation times. The ten-day in-phase and quadrature coefficients are $(0.866, 0.104)$. The unit, in-phase prediction lies within the

nominal delete-day regions for both baseline durations (Fig. 2b). The zero point lies inside the five-day region and outside the ten-day region. A separate joint two-parameter test, resampling six periods, gives $p = 0.25$ for zero and $p = 0.72$ for the unit, in-phase point with ten-day baselines. The phase estimates are compatible with the prediction, but the nominal delete-day ellipse alone does not establish a robust exclusion of zero. Three direct-potential alternatives to the solid-tide approximation change the lunar scale by less than 0.005 and the whole-template scale by less than 0.012 (Methods). Separate constituent scales give M2 0.94 ± 0.27 and N2 0.29 ± 1.08; N2 is not independently resolved.

Summer contributes 75.8% of the squared lunar template after projection against the other fitted columns, compared with 45.9% for the whole-template model without daily terms. These are design properties, not physical fractions of the tide. Daily terms also weaken the held-out diagnostic: three of six periods improve, with a pooled reduction of 0.467% (Fig. 1c).

A restricted wild bootstrap over six periods gives two-sided p-values of 0.0313 for zero whole-template response, both without and with ten-day daily terms. The five-day daily-term result is 0.0625. For M2+N2, the corresponding p-values are 0.0625 and 0.0938 for ten- and five-day blocks. With only 64 sign patterns and assumptions about period errors, these tests do not justify a unique high-significance detection claim for the lunar component. Numerical inversion of the ten-day period test gives a 95% acceptance interval of [0.51, 1.42] for the baseline-only whole template and [−1.18, 4.22] for M2+N2. These working sets express the small number of periods and differ from a Gaussian reading of the standard errors. Simulations of the actual error summaries, including their maximum, are reported in Methods. They do not cover arbitrary clock or transfer errors. Actual residual checks also expose limitations of a simple homogeneous Gaussian noise model, reinforcing the distinction between estimated response and calibrated detection strength (Methods).

The measured transfer control

The recorded round-trip monitor tests one part of the transfer system. We use the apparatus-designated B2 channel and pair it with clock measurements without interpolation. A regression bin is retained only when all its original members have unique valid monitor samples. Gross frequency excursions motivate a ±100-Hz screen around the recorded 10-MHz carrier, with unfiltered, ±10-Hz and ±1000-Hz alternatives retained. This screen changes only the monitor comparison's support.

At zero nominal time shift, the ±100-Hz criterion leaves 1,519 bins and 910,041 source members, all in summer 2026. On simultaneous valid second samples, the clock and nominal-optical-normalized round-trip channel have one-hour overlapping Allan deviations of 7.66×10^{-18} and 6.26×10^{-18} (Fig. 3a). The longest computed averaging time, 16,384 s, has only three contributing runs and a clock Allan deviation of 4.91×10^{-18}. It does not establish stability over the accumulated 20.1 sample days. No assumed factor of two converts this diagnostic into a calibrated one-way bound.

Adding B2 as a regressor changes the whole-template scale by 0.0057 ± 0.0073 without daily terms and 0.0415 ± 0.0345 with them; errors use paired day deletions. Figure 3b shows nominal 95% intervals and threshold sensitivity. Fixed ±5-s pairings are also retained. B2 itself has normalized template projections of 0.074 ± 0.087 without daily terms and 0.54 ± 0.46 with them, using delete-day errors. A small paired change in the clock scale therefore does not demonstrate a zero monitor projection. The result quantifies the tested linear association; clock effects and unmonitored laboratory branches remain possible contributors.

Longer periods and retained levels

For Mf, Mm, Msf and seasonal Sa and Ssa, the 2026 measurements retain their levels and share one global intercept (Fig. 4a). We fit all five quadrature pairs jointly with short-period tides and global daily terms. A single global drift is tested separately. No interval or calendar-block centres are adjusted for these fits.

The no-drift model has insufficient sensitivity to identify the predicted seasonal solar tide. Nominal delete-period 95% upper bounds on free-phase amplitude are $(1.60, 4.32, 2.70, 20.98, 11.11) \times 10^{-17}$ for Mf, Mm, Msf, Sa and Ssa (Fig. 4b). They are conditional statistical bounds: unknown clock corrections and residual calibration changes are not bounded. Adding a global drift expands the Sa and Ssa bounds to approximately 6.0×10^{-15} and 1.1×10^{-15}. Removing May, whose Yb correction is missing, also weakens seasonal constraints.

An independent calculation of the diurnally averaged solar potential gives a semiannual differential response near 2.02×10^{-20}. Similar station latitudes suppress this component. Even the no-drift Ssa upper bound exceeds that prediction by approximately 5,500; the Sa bound is about 75,000 times the current model amplitude. Calibrated levels alone would not close this sensitivity gap. The present sensitivity is far from this predicted signal, and the existing ocean-loading synthesis contains none of the five long-period constituents tested here. More elapsed time alone cannot substitute for calibrated levels and an adequate loading model.

Discussion

The Shanghai–Wuhan comparison supports a common response to the predicted tide across six separated periods from 20.1 nominal days of retained samples. Whole-template scaling, phase-sensitive subtraction and internal transfer of the amplitude support this interpretation within the stated models. Allowing solar-day terms retains a lunar response compatible with the prediction; joint phase tests and residual diagnostics limit component-specific attribution. Accumulated sample time quantifies the observations acquired, while phase coverage, calibration and noise structure determine their sensitivity. The measured monitor tests a specified transfer association while leaving clock and unmonitored-path contributions within the interpretation boundary.

The earlier ratio result and this analysis use overlapping observations for different targets (Table 1). The reported $0.87(15)$ and 5.7σ in the earlier, narrower model cannot be transferred to the present lunar estimate. More general nuisance and noise tests do not by themselves invalidate that earlier statement. The commissioning data are now kept separate because experimental configuration, not calendar length, defines which measurements can be combined. Unknown corrections and the retained-level plotting coordinate prevent direct interpretation of Fig. 4 as the calibrated absolute-ratio coordinate of the earlier summer analysis.

The European work establishes tidal tracking with a network, Kalman estimation and same-site controls.[5] Our independent comparison addresses consistent response over interrupted operation, separation from solar-day variation and the limits of retained-level inference. This is complementary experimental evidence, without claiming exclusive clock species, geographical scale or analysis techniques. Neither the whole-template scale nor this component decomposition independently estimates geophysical response parameters.

Dynamic chronometric sensing thus has more than one calibration timescale. Semidiurnal variations can be tested within observation blocks while response consistency is examined between periods. Seasonal sensing requires those relative levels themselves to remain calibrated. Improved clock and local-transfer monitoring would strengthen both short-period attribution and the search for slower potential changes.

References

1. Lisdat, C. et al. A clock network for geodesy and fundamental science. *Nature Communications* **7**, 12443 (2016). Source
2. Grotti, J. et al. Geodesy and metrology with a transportable optical clock. *Nature Physics* **14**, 437–441 (2018). Source
3. McGrew, W. F. et al. Atomic clock performance enabling geodesy below the centimetre level. *Nature* **564**, 87–90 (2018). Source
4. Hou, L. et al. Yb/Sr optical clock comparison at 2.2×10^{-18} uncertainty over a 1350-km fiber link. *arXiv*, 2610.34250v1 (2026). Source
5. Stahl, K. et al. Optical clock network tracks tidal variations of the gravity potential. *arXiv*, 2610.39635v1 (2026). Source
6. Ludlow, A. D., Boyd, M. M., Ye, J., Peik, E. & Schmidt, P. O. Optical atomic clocks. *Reviews of Modern Physics* **87**, 637–701 (2015). Source
7. Chou, C. W., Hume, D. B., Rosenband, T. & Wineland, D. J. Optical Clocks and Relativity. *Science* **329**, 1630–1633 (2010). Source
8. Bothwell, T. et al. Resolving the gravitational redshift across a millimetre-scale atomic sample. *Nature* **602**, 420–424 (2022). Source
9. Riehle, F. Optical clock networks. *Nature Photonics* **11**, 25–31 (2017). Source
10. Voigt, C., Denker, H. & Timmen, L. Time-variable gravity potential components for optical clock comparisons and the definition of international time scales. *Metrologia* **53**, 1365–1383 (2016). Source

11. Denker, H. et al. Geodetic methods to determine the relativistic redshift at the level of 10^{-18} in the context of international timescales: a review and practical results. *Journal of Geodesy* **92**, 487–516 (2018). Source
12. Jia, Z.-P. et al. Improved systematic evaluation of a strontium optical clock with uncertainty below 1×10^{-18}. *Metrologia* **63**, 025002 (2026). Source
13. Zhu, Q. et al. Ytterbium lattice clock with systematic uncertainty of 1.3×10^{-18} and instability at the 10^{-19} level. *arXiv*, 2606.10514v2 (2026). Source
14. Chen, F.-X. et al. 10^{-21}-Level optical frequency dissemination over 2067 km of noise-loaded field-deployed fiber network. *Light: Science & Applications* **15**, 276 (2026). Source
15. Schioppo, M. et al. Comparing ultrastable lasers at 7×10^{-17} fractional frequency instability through a 2220 km optical fibre network. *Nature Communications* **13**, 212 (2022). Source

Table 1

Table 1 | Related tidal estimates and their statistical targets. The estimates share observations and are not combined. The previous 2025–2026 component reduction is historical; the present common amplitudes use 2026 only, with 2025 commissioning records analysed separately. Values in parentheses or after ± denote the error scale specified in the last column; they are not a common calibrated confidence level.

Analysis	Observations and averaging	Tidal target and nuisance model	Scale	Error definition
Hou et al.[4]	Summer record; 16 intervals; non-overlapping 1,200-s inference windows	Whole predicted tide; interval-centred fits and precision weighting	0.87(15)	Reported weighted statistical error; reported 5.7σ under that analysis
Previous component analysis, v10	3,186 overlapping filtered outputs; 54 interval offsets	M2+N2; campaign-specific 24-h/12-h terms	0.86 ± 0.18	max(LS, delete-day SE)
Same v10 support, whole-template comparison	Identical 3,186 outputs, offsets and daily terms	Whole predicted tide; same offsets and daily terms	0.89 ± 0.20	max(LS, delete-day, delete-period SE)
Present 2026 reconstruction	2,905 non-overlapping 600-s means	Whole predicted tide; ten-day calendar baselines	0.88 ± 0.10	max(LS, delete-day, delete-period SE)
Present daily-term control	Same 2,905 means	Whole predicted tide; ten-day baselines and 24-h/12-h terms	0.79 ± 0.28	Same standard-error summary
Present lunar decomposition	Same 2,905 means	M2+N2; independent other tide groups and ten-day daily terms	0.89 ± 0.28	Same standard-error summary

Figure legends

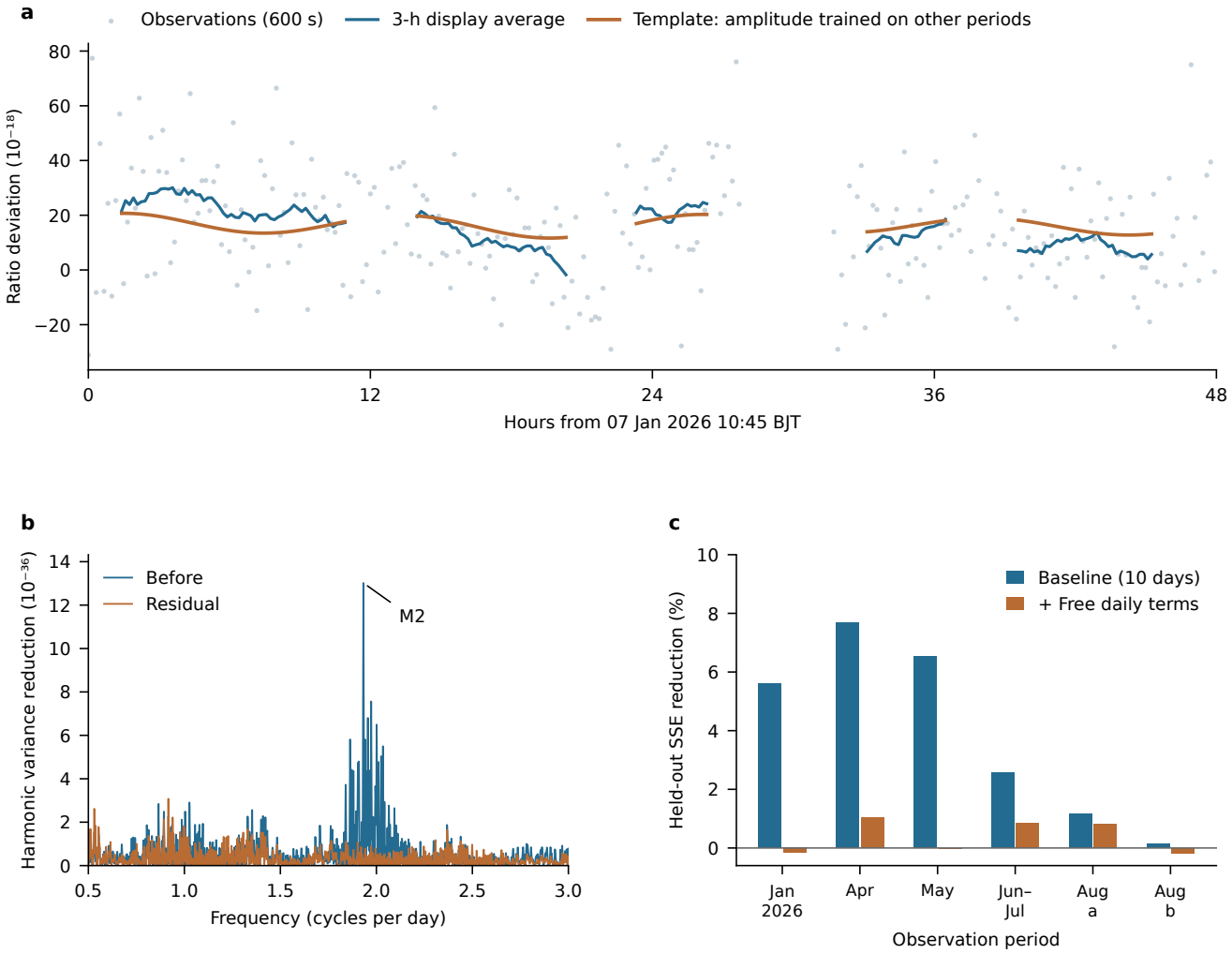


Fig. 1 | A common tidal-template response across interrupted observations. a, The 48-h window containing the largest number of retained source members (158,072 in 264 bins), selected without reference to fit quality. Points are 600-s averages after subtraction of one common reference used for the whole record. The blue curve averages 18 consecutive bins within uninterrupted supports for display. The orange curve applies the same display average to the template, with its amplitude trained on the other five periods and its calendar-block baselines fitted in the held-out period. No individual segment offsets are subtracted from the points. **b,** Actual-time harmonic scans of all 2,905 bins before and after whole-template subtraction, with the same ten-day baseline projection. The ordinate is harmonic variance reduction per observation, not a physical power spectral density or a significance threshold. **c,** Changes in held-out-period residual sums of squares when the amplitude is learned from the other periods. Blue and orange denote the baseline-only and free-daily-term designs. The held-out nuisance terms are fitted locally; the six folds share training data. All time-series points are repeated observations of the same apparatus, not independent experiments.

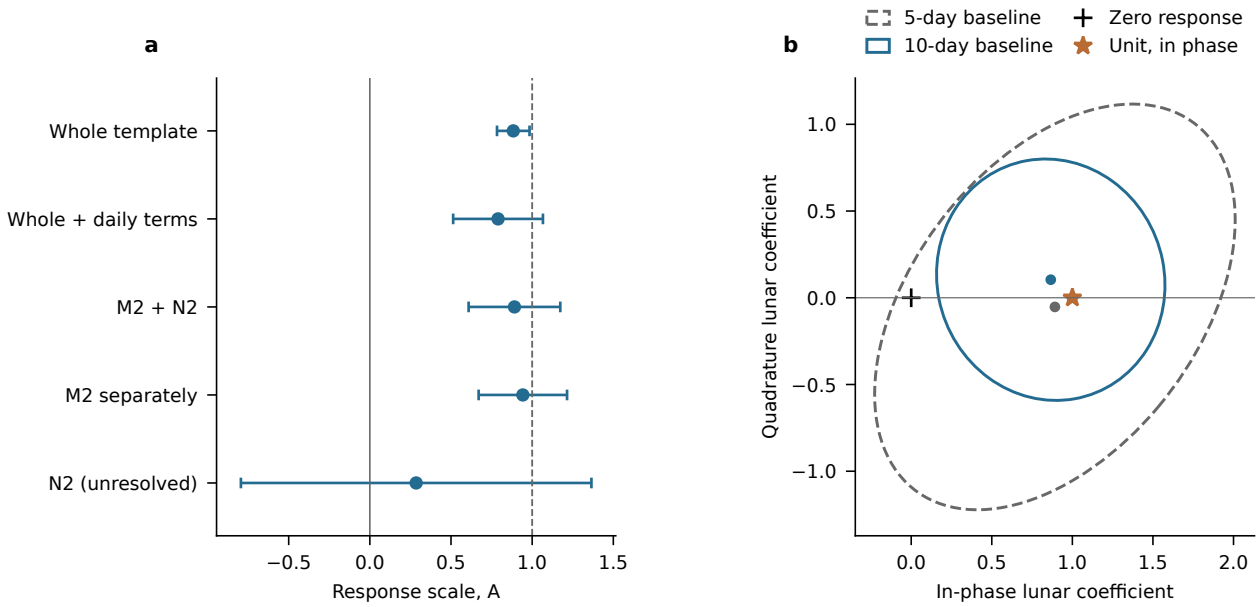


Fig. 2 | Lunar response after allowing solar-day variation. **a,** Scales estimated from the same 2,905 averages. All rows except the first include ten-day 24-h and 12-h nuisance terms. M2 and N2 are estimated jointly with independent scales in the last two rows. Bars show one standard-error summary, max(LS, delete-day, delete-period); the vertical references mark zero and unit response. The rows are correlated estimates of different targets. **b,** In-phase and quadrature coefficients of a common lunar response for five-day and ten-day baselines. Curves are nominal 95% delete-day working-model regions for the two coefficients; points mark their estimates. The cross and star denote zero and the unit, in-phase prediction. Zero is inside the five-day region and outside the ten-day region; the unit, in-phase point lies within both. No observation times or phases are adjusted. For the same two-parameter target, the six-period wild test gives ten-day $p = 0.25$ at zero and $p = 0.72$ at the unit point (Methods, Table 5); it uses different covariance estimation and calibration from these ellipses. Neither procedure corrects for exploratory model selection or arbitrary noise.

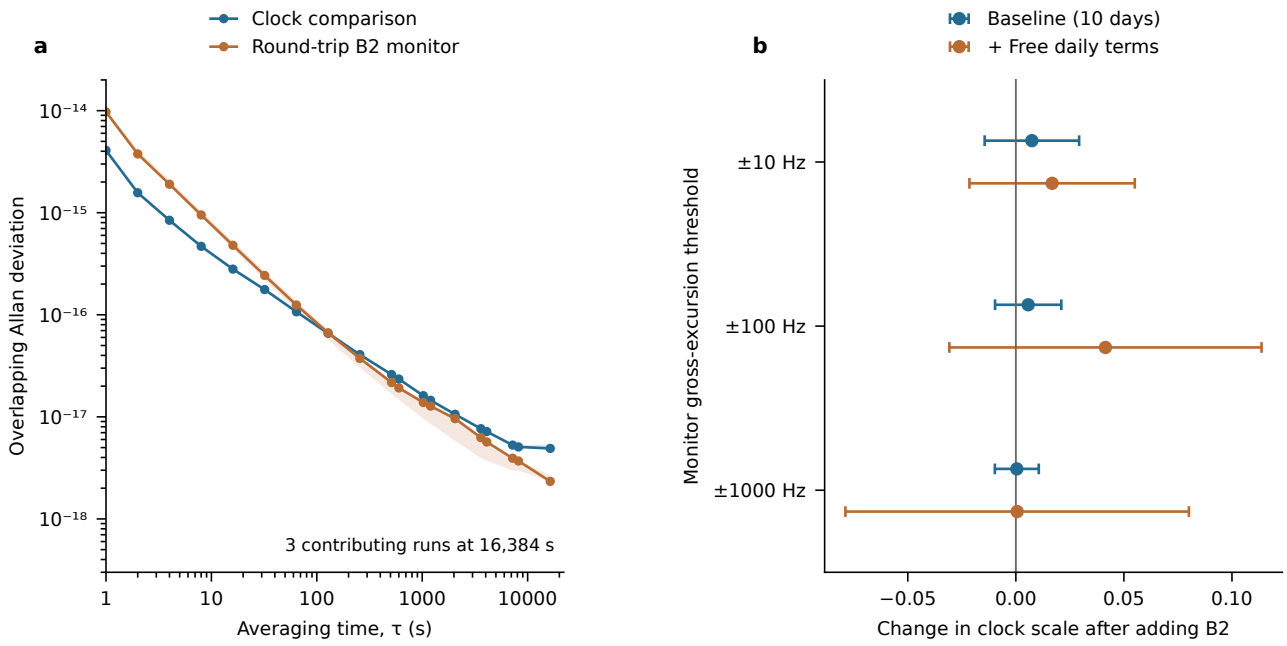


Fig. 3 | Scope of the measured round-trip transfer control. **a,** Overlapping Allan deviations of the clock comparison and predesignated B2 monitor on simultaneous valid second samples. Solid curves use the ±100-Hz gross-excursion criterion (913,692 common seconds before averaging-time run requirements). Shading spans the ±10-Hz and ±1000-Hz sensitivity results as well as the main curve; it is not a confidence band. No differences cross gaps, and contributing runs must span at least four averaging times. At 16,384 s, only three runs contribute 268,673 overlapping differences, which are not independent replicates. The monitor is normalized by the nominal optical carrier without an assumed one-way conversion. **b,** Paired changes in the clock's whole-template scale on adding B2, at zero nominal pairing shift. Thresholds leave 1,476, 1,519 and 1,520 complete 600-s bins. Error bars are nominal 95% delete-day intervals for the paired changes, with 19 degrees of freedom; they are not interval differences computed from independent fits. Baseline-only and free-daily-term designs use the same support within each threshold. The control covers the recorded channel and time overlap, including neither unmonitored laboratory branches nor the complete year.

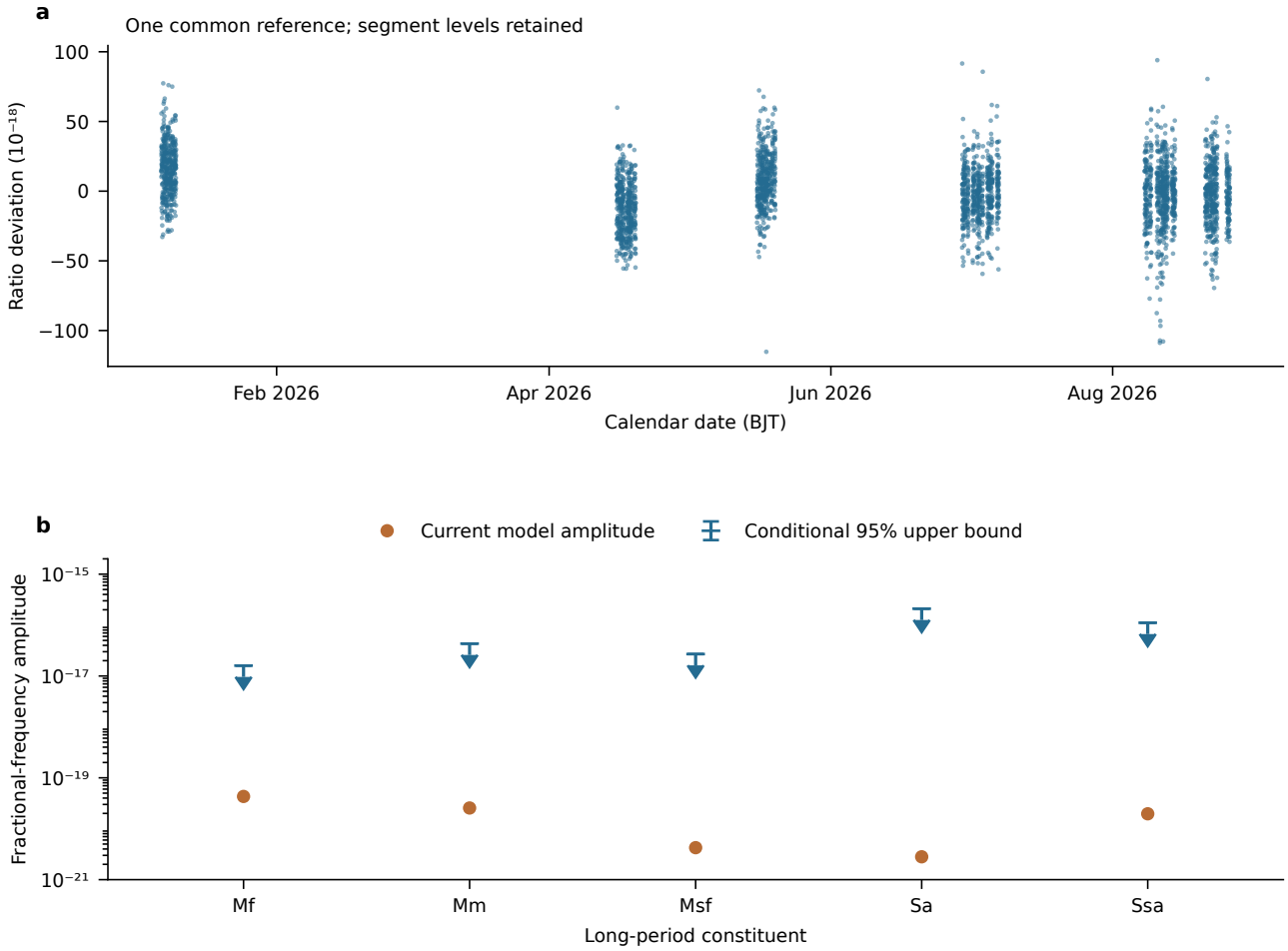


Fig. 4 | Preserved frequency levels and long-period sensitivity. **a,** All 2,905 reconstructed 2026 ratio averages, with one common plotting reference subtracted. No fitted segment or period centres are removed. Points retain the actual gaps and period-to-period level differences. **b,** Nominal 95% upper bounds on free-phase amplitudes from the 2026-only, single-intercept model with joint short-period and global daily terms, without a global drift. Caps indicate upper bounds and arrows indicate the permitted lower-amplitude direction; arrow lengths are graphical. Bounds use the largest eigenvalue of each two-coefficient delete-period covariance and an approximate two-dimensional region with five denominator degrees of freedom. They are individual bounds, conditional on the stated calibration and noise assumptions. Orange points give amplitudes from the current theoretical decomposition, not measured signals. The five-constituent family bounds and drift sensitivity are specified in Methods. Unknown calibration terms and incomplete long-period loading prevent these bounds from being interpreted as unconditional gravitational limits.

Methods

Coordinate and frequency conversion

The principal inference is restricted to 2026. Frequency conversion follows the separately recorded year configurations for the Shanghai Sr and Wuhan Yb comparison.[4] The 2025 conversion is retained only for its separate commissioning-stage analyses. With $N_{1397} = 858456$, $N_{1550} = 773598$, $N_{1156} = 1295739$ and $N_{1550,\mathrm{WH}} = 966996$, the 2025 chain gives

$$f_{1397} = \frac{f_{\mathrm{Sr}}}{2(1-\alpha)}, \quad \alpha = \frac{10.93629485}{25N_{1397}}, \quad f_{1550,\mathrm{WH}} = \left(N_{1550} + \frac{8}{25}\right)\frac{f_{1397}}{N_{1397}}.$$

The eight Shanghai reference units include the seven in the transmitted carrier and the additional unit at Wuhan. The Wuhan repetition frequency is $f_{\mathrm{Yb}}/(2N_{1156})$, and its 10-MHz reference is $f_{\mathrm{Yb}}/(40N_{1156})$. The comb offset is 26 such reference units. If the recorded counter value q is in nominal hertz realized by this Wuhan reference, the locked ratio is

$$R_{\mathrm{lock}} = \frac{N_{1156}}{N_{1397}} \frac{N_{1550} + 8/25}{N_{1550,\mathrm{WH}} + 26/20 + dq/(20\times 10^7)} \frac{1}{1-\alpha}.$$

The reconstruction uses $d = 2$ and the stated α in 2025, and the separately recorded 2026 convention $d = 1$, $\alpha = 0$. A 2025 counter reading near 66 MHz corresponds to a beat near 132 MHz before electronic division; the recorded 2026 configuration is near 33.6 MHz. The counter-reference convention is an explicit experimental assumption, not a consequence of a reference being labelled 10 MHz. The electronic divider is restored once and is unrelated to a round-trip fibre correction.

With signed clock shifts a for Sr and b for Yb, we form $R = R_{\mathrm{lock}}(1+a)/(1+b)\,(1+\Delta G)$, with the inherited constant $\Delta G = -3.116\times 10^{-15}$. The coordinate is $y = (R/R_{\mathrm{ref}} - 1)10^{18}$, with $R_{\mathrm{ref}} = 1.2075070393433377166$. Decimal anchors prevent loss of small ratio changes through subtraction in double precision: the small deviation is not formed by subtracting the two full ratios in float64. Positive y denotes a faster Wuhan Yb clock relative to Shanghai Sr after the stated corrections. For positive Newtonian potential U, the clock-rate contribution is $-\Delta U/c^2$, with $\Delta U = U_{\mathrm{WUH}} - U_{\mathrm{SHA}}$; this fixes the tidal sign below. The constant potential correction and ratio anchor do not constitute a new static-geopotential or absolute-ratio measurement.

The eight periods are September and October 2025; January, April and May 2026; June–July 2026; and early and late August 2026. In units of 10^{-18}, the corrected Sr density entries are −0.43, −0.41, −0.30, −0.33, −0.25, −0.24, −0.33 and −0.29. In the same units, the first six Zeeman corrections are −178.50, −178.87, −179.14, −180.19, −180.18 and −178.57; August Zeeman and all blackbody corrections are recorded as online. The background-gas entry is -0.67×10^{-18}. AC entries for the first five periods are missing; the final three are $[7.44, 8.93, 9.03]\times 10^{-18}$. The documented positive 37.2-mHz feedback offset is applied to the first three periods at the assumed 698-nm optical-frequency node, giving 86.6119×10^{-18} using the nominal wavelength. The node interpretation is retained as a condition. Printed Yb shifts for original groups 1–9 are used for the principal reconstruction, with the component sums examined separately. The May Yb correction is missing. Unknown entries represented as zero in the computational coordinate are not measured zero shifts. The 86.6-unit feedback contribution exceeds the roughly ten-unit tidal peak-to-peak scale; unknown AC and May Yb terms have no certified amplitude bound. The table is frozen for these refits, but the record does not establish that every correction preceded all earlier tidal-residual inspection. Nor is constancy within a block independently certified: constant contributions are absorbed by its baseline, whereas time-varying terms may project onto the tide.

Commissioning records and corrections

Reported commissioning, processing changes and unresolved offsets motivate separate September and October 2025 stage- or interval-intercept analyses, without shared amplitudes across months or with 2026. This scope was clarified during reanalysis.

An earlier uncompensated Yb configuration shift of approximately -1.1×10^{-17} in September–October 2025 may overlap the updated totals of -9.7 and -9.5×10^{-18}, which include a lattice entry of -8×10^{-18}.

Inclusion remains unverified, so the older shift is not added again. An additional shift δb would multiply the ratio by $(1+b)/(1+b+\delta b)$; $\delta b = -1.1 \times 10^{-17}$ would increase it by approximately 1.1×10^{-17}.

Adding 11×10^{-18} verifies the additive-invariance property of complete intercept designs; it does not certify the actual multiplicative correction or its provenance. With only two observed days per 2025 month, day/period deletions cannot establish correlated-noise coverage. Stage whole-template fits give September 0.87 ± 0.67 and October 1.57 ± 0.26 (LS errors only), with substantially larger errors after daily controls. These commissioning diagnostics are not calibrated independent detections.

Observation selection and averaging

The corrected experimental intervals select 1,966,912 unique source rows from 42 files, retaining fractional-second BJT labels. Intervals include the first displayed PDF second and exclude the second immediately after the last displayed second. Rows are sorted by time, file and source row and deduplicated by source identity. Finite beat offsets below 1 MHz in magnitude and strictly within 10 Hz of the plausible-sample median leave 1,966,908 rows. The three-row difference from the experimental table's count is retained with boundary provenance.

Within each effective interval, 300-s block means are compared with a 13-block running median using reflected boundaries. The median absolute deviation of the block residuals, divided by 0.6744897501960817, defines a screening scale. Blocks exceeding six scales in absolute residual and their immediate neighbours are excluded. The local residual is used only for selection; retained beat values are not median-subtracted. No fitted tide residual enters the screen. It removes 11,131 additional rows. Non-overlapping 600-s means require at least 300 retained members; 3,266 means contain 1,952,992 members. The 2026 restriction retains 2,905 of these means and 1,737,073 members; September and October 2025 retain 160 and 201 means, respectively. Tidal and daily-harmonic columns are averaged over their exact members. Baseline block indicators are assigned by the bin-mean timestamp, retaining a complete bin when its members cross a block boundary. Observations are not interpolated. Day deletions use bin-mean BJT days; the parent 3,266-bin record includes 16 midnight-crossing bins, so this is not deletion of every raw sample from the named civil day. The 2026 retained source members occupy every local hour; 45.6% fall between 06:00 and 18:00 BJT. Table 3 records occupied ten-day blocks. All use the recorded 2026 conversion, so none contains the year-convention change; within-block drift tests do not identify a particular hardware event.

Tidal prediction and component design

The common prediction combines solid-Earth and ocean-loading series on a 90-s grid. The solid-Earth contribution uses pysolid displacements under the recorded IERS 2010 implementation,[16] converted as

$$s(t) = -\frac{1 + k_2 - h_2}{h_2} \frac{g[u_{\mathrm{WUH}}(t) - u_{\mathrm{SHA}}(t)]}{c^2},$$

with $h_2 = 0.6078$, $k_2 = 0.3010$ and $g = 9.794\,\mathrm{m\,s^{-2}}$. This is a displacement-based approximation, not a complete independent degree-dependent potential calculation. Ocean loading uses 18 constituents from grids locally documented as FES2022b.[17] The FES family uses hydrodynamic modelling and data assimilation; the FES2014 atlas paper describes that earlier version, not a validation of FES2022.[18] The FES2022 Tide product was funded by CNES, produced by LEGOS, NOVELTIS and CLS and made freely available by AVISO. Our loading series is obtained by convolution with PREM load-Love-number Green functions containing $1 + k'_n - h'_n$. The archived Han–Wahr table[19] through degree 696 is extended to degree 40000, with a cosine taper over the final 30% and the recorded centre-of-solid-Earth degree-one convention. It includes direct water potential, induced potential and displacement. On the identical member-matched 2026 bins, globally demeaned solid, ocean-loading and total RMS values are $(3.245, 1.507, 4.410) \times 10^{-18}$. Loading has 34.2% of the total RMS, with solid–loading correlation 0.680; squared RMS values are not additive because of their covariance. Atmospheric loading is not modelled or bounded here. Free exact 24-h/12-h terms control corresponding empirical variation, but do not certify removal of a physical atmosphere-loading field. Local surface-pressure differences cannot supply that certification.

The 360,001 theory samples are projected onto quadratures of Q1, O1, P1, K1, N2, M2, S2, K2, Mf, Mm, Msf, Sa and Ssa plus a constant (rank 27), retaining the remainder. BJT is converted to UTC once,

using the saved FES convention and TT−UTC = 69.184 s. Whole-template fits multiply the complete prediction; component fits subtract the fixed remainder once.

In fractional-frequency units of 10^{-18}, the lunar design is

$$y - F = A_L C_L + a_{12} C_{12} + a_D C_D + Z_b \gamma + Z_{24,12} \beta + \epsilon.$$

C_L contains M2+N2, C_{12} contains S2+K2, and C_D contains the four diurnal constituents. F contains the rest of the theory, including the long-period terms. The daily design has sine and cosine columns at one and two cycles per day in each occupied block. The 2026 ten-day lunar model has 53 parameters and full rank. The blocks are fixed by $\lfloor t/(10 \times 86400) \rfloor$ on the BJT-labelled epoch coordinate; five-day sensitivity uses the analogous rule. These are calendar blocks, irrespective of actual uptime. The S2+K2 scale is not interpreted as separately resolved from the exact 12-h nuisance terms. The exploratory slope test adds one linear column in each occupied block, centred on that block's observed mean time and divided by its fixed duration. All six constant-versus-slope comparisons retain identical data and templates. Paired deletion differences estimate the uncertainty of each change in scale. These short-period baselines and slopes are not used for long-period inference. Further user-requested 20-day, 30-day and per-observation-period controls give lunar scales 0.867 ± 0.246, 0.796 ± 0.202 and 0.761 ± 0.205, respectively. They retain the same 20.1 days of sample time; the longer constraints reduce nuisance freedom rather than add observations. The 30-day grid shares one baseline and daily pattern across the two August periods, whereas the per-period design keeps them separate. All three have six-period wild $p = 0.03125$, whereas the five-/ten-day results are weaker. These post-review sensitivities are not used to select a more significant primary baseline.

Three theory sensitivities replace only the solid-Earth contribution with direct geometric Sun–Moon potentials from DE430:[20] degree two with constant response $1 + k_2 - h_2$; degrees two and three with their respective constant responses; and degrees two and three with nominal order-dependent real k_{nm} from IERS Table 6.3.[16] The third-degree displacement response is $h_3 = 0.292$. Fixed station coordinates are transformed using IAU 2006/2000A rotation, with UT1 approximated by UTC and zero polar motion. These controlled approximations omit full frequency-dependent and complex responses and are not replacements for a complete IERS potential implementation. On the identical 2,905 member-matched 2026 observations, the alternatives give whole-template scales 0.876, 0.872 and 0.872, and lunar scales 0.895, 0.887 and 0.887. All alternative results are retained; the primary template is unchanged. Neither these variants nor their small scale shifts bound the missing frequency-dependent complex Love responses. Using the alternative coordinates reported by Hou et al.[4] (Wuhan 30°31′ N, 114°19′ E, 30 m; Shanghai 31°14′ N, 121°29′ E, 55 m), while holding the same direct-potential algorithm and ocean loading fixed, changes A_T by approximately +0.0066 and A_L by +0.0045. The coordinate-only solid-template difference has RMS 0.043×10^{-18}. This is a controlled sensitivity, not certification of either coordinate set at the actual clock transition point; their height references and clock-point binding require confirmation.

Estimation, correlation and validation

Column-normalized SVD fits retain residual-scaled LS, delete-day and delete-period covariance, plus elapsed-time Bartlett HAC sensitivities at one-, three- and ten-day bandwidths. The jackknife principle[21] gives $C_{\mathrm{JK}} = (G-1)G^{-1} \sum_g (\hat{\theta}_{-g} - \bar{\theta})(\hat{\theta}_{-g} - \bar{\theta})^T$; independence of deletion groups remains an assumption. Unsupported nuisance columns are removed; target loss and rank failures are recorded. The largest LS/day/period SE is a scalar error summary, not a joint covariance.

The restricted wild bootstrap fits each null, flips residuals by a common ±1 within each period, refits and studentizes with delete-period SE.[22] All 64 patterns are enumerated. Period independence and symmetric errors are assumptions; nuisance-residual estimation prevents exact randomization inference. The minimum two-sided p-value is 0.03125. Exploratory choices are not multiplicity-corrected. For the joint in-phase/quadrature target, the restricted residual is $M_Z(y - Lb_0)$, with both target columns excluded from Z. Each sign pattern refits both coefficients and the full delete-period covariance, using its studentized quadratic form. Table 5 retains all five baseline designs. Direct QR refits verify the ten-day tests; Cholesky checks cover all 640 observed bootstrap covariances. Separate fixed-design simulations (1,024 per design/noise scenario; seed 202611053) show low power in several scenarios, so non-rejection does not establish either null point.

Fixed-design coverage is evaluated for all six five-/ten-day whole-template, daily-term and lunar designs, with 1,024 simulations in each of five scenarios. They are independent Gaussian errors with variance 500 in working 10^{-18} units; white noise of SD 10 plus an OU process of SD 20 and time constant 1 h, 12 h or 3 days; and white noise plus independent period constants of SD 30. Period constants may be absorbed by the nuisance design, so this last scenario is not a general drift test. Actual times and design matrices are retained, with seed 202611052 and distinct scenario streams. Table 2 reports bias, mean standard errors and 95% coverage for the two ten-day targets. The maximum-SE rule is explicitly tested: $\hat{A}\pm u_{\max}$ is a one-error summary, whereas $\hat{A}\pm 1.96u_{\max}$ and $\hat{A}\pm t_{5,0.975}u_{\max}$ are separate nominal 95% constructions. The latter gives 98.1–99.6% coverage for the whole template and 99.4–100% for the lunar target across these scenarios. None is certified under unknown systematics or exploratory model selection.

Wild coverage records acceptance of simulated true values, separately from numerical set construction. Inversion partitions the real line at bootstrap-t polynomial roots and checks root membership, including isolated points. Tangent-root tests and independent matrix calculations agree; no isolated accepted points occur here. The ten-day daily-term whole-template 95% set is disconnected: $[-0.541, -0.294] \cup [0.241, 0.367] \cup [0.385, 2.568]$. Its convex hull is not substituted for the actual set.

The historical 124/200 injection coverage concerns one SE, not 95%; its Wilson interval includes 68.27%. It does not certify extreme-tail significance. As an additional noise check, a homogeneous Gaussian white-plus-OU model averages its exponential covariance over every actual retained member pair and uses white-bin variance proportional to $600/n_i$. ML and REML search the same 71-point grid. In all four ten-day mean designs, fitted whitened residuals have lag-one correlations between -0.13 and -0.17, period RMS ratios of 1.37–1.38 and kurtosis of 4.82–5.12. None of 4,096 conditional simulations per fitted model exceeds the observed maximum absolute correlation over lags 1–12, RMS ratio or kurtosis. Simulations retain the full mean projection but fix the selected covariance; their tail proportions are not selection-adjusted goodness-of-fit probabilities. This noise family is inadequate for those diagnostics. Its likelihood significance is therefore not used to support detection, and the physical source of the discrepancy remains unidentified.

Exposure and conditional sensitivity

Chronological prefixes first reach 2, 4, 8, 12 and 16 nominal sample days, plus full support, retaining complete bins (Table 4). Thresholds preceded this calculation but followed full-data exploration. Nested prefixes change phase coverage, period composition and nuisance columns; they are not independent replications or sequential tests. LS, day and period errors are separate; fewer than three deletion groups are marked unavailable. Precision need not improve monotonically.

A conditional technical benchmark uses the measured summer 600-s clock Allan deviation, $\sigma_{600} = 2.3476\times10^{-17}$, as a white-frequency reference. Independent bin variances $v_i = \sigma_{600}^2(600/n_i)$ propagate through the unchanged OLS target operator O as $O\,\mathrm{diag}(v_i)O^T$. This does not reweight observations or certify independence and equal noise across periods. The full-support benchmark errors are 0.101 and 0.268 for whole-template and lunar scales. At four nominal days, the lunar benchmark error is 4.33 times that obtained by scaling the full-support error as $T^{-1/2}$, reflecting the different sampling and nuisance geometry. No Allan extrapolation to 20 days or universal detection time is inferred.

Spectral and held-out diagnostics

Harmonic scans generalize floating-mean least squares[23] using the nuisance projector $M_Z = I - ZZ^+$. At each frequency, the two-quadrature fit to $M_Z y$ gives residual-sum-of-squares reduction divided by sample size. Actual bin-mean times scan 0.25–4 cycles/day at $1/(5\Delta t)$ spacing, including specified constituents. Subtracting $\hat{A}_T M_Z T$ leaves the same projection. Gaps remain intact. This is neither a physical PSD nor a calibrated frequency-search significance test.

For leave-period-out diagnostics, five periods train the amplitude. The test response and template are projected against locally fitted nuisance columns before comparing residual sums of squares. The test amplitude is fixed; training observations overlap across folds.

Fibre monitor and stability

The 29 recorded monitor files are interpreted according to the experimental B2 designation. Their BJT-labelled nominal one-second axes use the floored first file timestamp plus row index, matched to the corresponding nominal clock axes; repeated or nonfinite monitor seconds are excluded. Fixed pairings at −5, 0 and +5 s test sensitivity to that convention, without certifying hardware synchronization or logging latency. Complete-bin matching retains all or none of a clock bin's original members. At zero pairing, the unfiltered common support contains 915,039 second samples; 1,347 have B2 deviations exceeding 100 Hz. The ±100-Hz criterion is a post-inspection gross-excursion screen, with ±10 and ±1000 Hz alternatives retained, rather than a threshold chosen for tidal agreement.

We compare $y = A_T T + Z\eta + \epsilon$ with $y = A'_T T + Z\eta' + \lambda B2 + \epsilon'$. Paired day deletions refit both models and estimate the uncertainty of $A'_T - A_T$ from the same deletions. The monitor's own template projection is also retained. Its frequency deviations are normalized by the nominal carrier 193399200000000 Hz only for a round-trip diagnostic. No division by two or normalization by the 10-MHz RF carrier is used to assert a one-way fractional optical-frequency error.

Overlapping Allan deviations[24] compare adjacent τ-length means on contiguous common seconds; no term crosses a gap, and runs shorter than 4τ are omitted. Term counts are retained. Summer monitoring is not extrapolated to other periods. Timestamp labels and ±5-s pairing checks do not certify counter synchronization. No same-site null comparison is available.

Long-period amplitudes and upper bounds

Long-period models jointly fit five quadrature pairs, short-period group scales and global daily terms on 2026 data with one intercept. Alternatives add one global drift, use component-sum corrections or omit May. No segment, period or month intercepts are used; all 2025 measurements are excluded.

For a pair $\hat{b}$ with covariance C and denominator degrees of freedom ν, a conservative radial bound from the nominal ellipse is

$$U = \|\hat{b}\| + \sqrt{2F_{2,\nu}(1-\alpha)\lambda_{\max}(C)}.$$

We use $\alpha = 0.05$ for an individual bound and 0.01 for the five-target Bonferroni family. With six period clusters, $\nu = 5$ is a working approximation. The no-drift family upper bounds for Mf, Mm, Msf, Sa and Ssa are $(2.30, 6.18, 3.75, 30.38, 15.28) \times 10^{-17}$. Omitting May leaves five periods and gives individual Sa and Ssa bounds of approximately 1.37×10^{-15} and 9.36×10^{-16}. Bounds are conditional on the calibration coordinate; missing early Sr AC corrections and other unbounded changes prevent an unconditional gravitational limit.

The independent solar check averages the degree-two potential over hour angle. With P_2 the second Legendre polynomial, it evaluates $-(1+k_2-h_2)GM_\odot[r_W^2 P_2(\sin\varphi_W) - r_S^2 P_2(\sin\varphi_S)]P_2(\sin\delta)/(r_\odot^3 c^2)$ using DE430[20] and the archived theory-input coordinates (Wuhan: 30.529° N, 114.357° E, 50 m; Shanghai: 31.130° N, 121.570° E, 20 m), converted to geocentric radii and latitudes for this calculation. This checks the small Sa/Ssa scale without fitting observations. It does not provide a complete seasonal-loading model. The existing 18-constituent loading series contains none of Mf, Mm, Msf, Sa or Ssa.

Computational reproducibility

Independent Shanghai–Wuhan acquisition is reported by the experimental team. The new template, spectral and component analyses followed access to the European work and critique. Route length, clock species and geographical configuration are not claimed as exclusive experimental categories.

Files, exclusions, source-member mappings, derived data, designs, covariances, seeds and figure inputs are versioned. Independent Decimal/datetime readback verified hashes and row counts for all 36 contributing files and sampled 3,600 records: timestamps and row identities agree, with maximum beat-offset difference 1.82×10^{-12} Hz. This is sampled verification, not a complete second reduction. All 1,737,073 retained 2026 source identities are unique and assigned to one bin; re-averaging archived pointwise ratios agrees within 2.2×10^{-13} in working 10^{-18} units. QR/FWL checks agree and source

hashes are unchanged. Principal calculations use Python 3.12.3, NumPy 2.5.4 and SciPy 1.18.1; astronomy uses NumPy 2.5.3 and pyTMD 3.0.9. Matplotlib renders the figures. Numerical reproducibility does not establish calibration or coverage.

Data availability

Versioned manifests, derived series, exclusions and figure inputs are retained locally. Authors must specify access and restrictions before submission; no public deposit is claimed.

Code availability

Versioned analysis, validation and plotting code includes source hashes and execution records. Authors must confirm access for editors, referees and readers; no public code release is claimed.

Table 2 | Fixed-design simulation performance. Each row uses 1,024 simulated responses. T: ten-day whole template without daily terms; L: M2+N2 with daily terms. **a,** Bias and mean standard errors, in dimensionless scale units. **b,** Nominal 95% coverage (%): individual LS, delete-day and delete-period t intervals; $1.96u_{\max}$; and acceptance of the true value by the wild test. Monte Carlo uncertainty is retained locally. These are simulated data, not additional observations.

a. Bias and standard errors

Target / noise	Bias	LS SE	Day SE	Period SE
T / iid	0.000	0.096	0.096	0.091
T / OU 1h	-0.007	0.094	0.264	0.255
T / OU 12h	-0.002	0.084	0.247	0.226
T / OU 3d	+0.003	0.064	0.146	0.134
T / period constants	+0.001	0.043	0.043	0.041
L / iid	-0.001	0.255	0.298	0.270
L / OU 1h	-0.002	0.238	0.799	0.714
L / OU 12h	+0.031	0.207	0.837	0.631
L / OU 3d	+0.004	0.164	0.594	0.359
L / period constants	+0.004	0.114	0.134	0.122

b. Interval coverage (%)

Target / noise	LS	Day	Period	Max-SE	Wild
T / iid	95.0	94.8	94.2	96.7	96.5
T / OU 1h	49.9	94.1	93.7	94.5	96.5
T / OU 12h	49.7	95.6	94.1	96.1	97.4
T / OU 3d	60.9	96.4	94.0	95.9	96.9
T / period constants	94.7	93.9	93.8	96.0	96.0
L / iid	95.2	96.8	94.4	98.3	96.9
L / OU 1h	50.2	96.3	93.1	96.9	96.7
L / OU 12h	49.9	99.0	92.9	98.8	96.5
L / OU 3d	67.0	99.8	93.5	99.8	95.8
L / period constants	96.2	97.5	94.1	98.3	97.5

Table 3 | Ten-day block support in 2026. Assignment uses the mean bin time; boundary-crossing source members remain intact. The start is inclusive and the end exclusive. All dates are BJT in 2026. Block duration describes the model, not continuous uptime.

Block	Calendar start–end	Bins	Source members	Observation period
2046	01-07–01-17	403	241,474	January
2055	04-07–04-17	181	108,008	April
2056	04-17–04-27	259	153,774	April

Block	Calendar start–end	Bins	Source members	Observation period
2058	05-07–05-17	121	72,085	May
2059	05-17–05-27	299	178,683	May
2063	06-26–07-06	527	315,603	June–July
2064	07-06–07-16	72	42,708	June–July
2067	08-05–08-15	592	355,036	August a
2068	08-15–08-25	341	203,702	August b
2069	08-25–09-04	110	66,000	August b

Table 4 | Accumulation on the actual 2026 record. Sample days sum retained members at a nominal one-second gate; span is elapsed BJT time. T and L denote the whole-template and daily-controlled lunar targets. **a,** Support and conditional independent-bin benchmark standard errors. **b,** Observed standard errors by method. "—" means fewer than three deletion groups; no missing error is assigned zero. These pointwise, nested comparisons do not have simultaneous confidence coverage or an optional-stopping guarantee.

a. Exposure, support and conditional benchmark

Sample days	Span (d)	Days / periods	T benchmark	L benchmark
2.003	2.2	3 / 1	0.362	2.985
4.007	99.9	6 / 2	0.241	2.599
8.001	131.7	12 / 3	0.143	0.911
12.005	179.6	20 / 4	0.123	0.424
16.004	217.8	28 / 5	0.107	0.314
20.105	231.3	35 / 6	0.101	0.268

b. Observed scale standard errors

Sample days	T LS	T day	T period	L LS	L day	L period
2.003	0.296	0.628	—	2.401	8.034	—
4.007	0.190	0.241	—	2.031	2.952	—
8.001	0.114	0.093	0.112	0.720	0.985	1.025
12.005	0.098	0.083	0.072	0.337	0.427	0.413
16.004	0.094	0.122	0.125	0.275	0.338	0.205
20.105	0.088	0.100	0.097	0.234	0.282	0.188

Table 5 | Joint lunar phase tests on the same observations. Six-period restricted wild tests compare the in-phase/quadrature pair with zero and with the unit, in-phase prediction. All 64 sign patterns are retained and each is studentized by its own delete-period covariance. These exploratory tests differ from fixed-phase scalar tests and the nominal delete-day ellipses in Fig. 2b. All designs use the same 2,905 observations; the primary ten-day baseline is unchanged.

Baseline	In phase	Quadrature	p, zero	p, unit in phase
5 days	0.892	-0.053	0.3125	0.9062
10 days	0.866	0.104	0.2500	0.7188
20 days	0.867	0.002	0.0312	0.6250
30 days	0.794	0.099	0.1250	0.2500
Observation period	0.754	0.067	0.0312	0.1250

Methods references

16. Petit, G. & Luzum, B. (eds) IERS Conventions (2010). *IERS Technical Note* **36** (2010). Source
17. LEGOS, NOVELTIS & CLS FES2022 (Finite Element Solution) Ocean Tide. *CNES data set* (2024). Source

18. Lyard, F. H., Allain, D. J., Cancet, M., Carrère, L. & Picot, N. FES2014 global ocean tide atlas: design and performance. *Ocean Science* **17**, 615–649 (2021). Source
19. Han, D. & Wahr, J. The viscoelastic relaxation of a realistically stratified earth, and a further analysis of postglacial rebound. *Geophysical Journal International* **120**, 287–311 (1995). Source
20. Folkner, W. M., Williams, J. G., Boggs, D. H., Park, R. S. & Kuchynka, P. The planetary and lunar ephemerides DE430 and DE431. *Interplanetary Network Progress Report* **42-196**, 1–81 (2014). Source
21. Efron, B. & Stein, C. The Jackknife Estimate of Variance. *The Annals of Statistics* **9**, 586–596 (1981). Source
22. Cameron, A. C., Gelbach, J. B. & Miller, D. L. Bootstrap-Based Improvements for Inference with Clustered Errors. *Review of Economics and Statistics* **90**, 414–427 (2008). Source
23. Zechmeister, M. & Kürster, M. The generalised Lomb-Scargle periodogram: a new formalism for the floating-mean and Keplerian periodograms. *Astronomy & Astrophysics* **496**, 577–584 (2009). Source
24. Riley, W. J. Handbook of frequency stability analysis. *NIST Special Publication* **1065** (2008). Source

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